Energy-Oriented CGLA Mapping of a Memory-Polynomial Digital Predistortion Kernel

Takuto Ando and Yasuhiko Nakashima
Nara Institute of Science and Technology, Ikoma, Nara, Japan

APCCAS 2026 · Epapers ID 2103

Background & Objective

Digital Predistortion

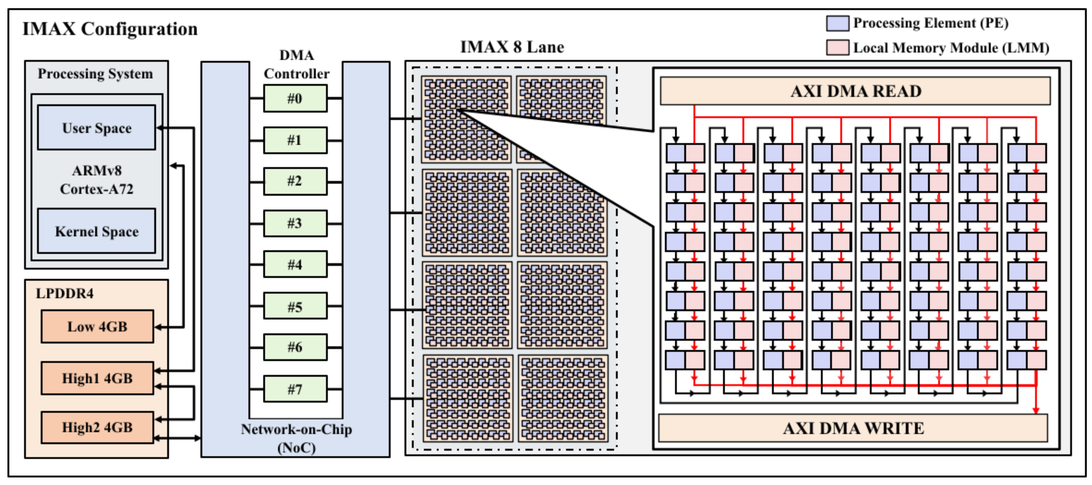
- DPD compensates nonlinear RF-transmitter distortion.
- A memory-polynomial form reuses coefficients and input history.

Objective

Can a 15-term complex-MAC reduction map to a one-dimensional PE/LMM stream?

The host generates basis terms within end-to-end timing. IMAX executes the reduction. RF hardware and adaptation are outside scope.

CGLA Accelerator IMAX



CPU-Grounded Linear Array (CGLA)
Programmable ASIC architecture with a PE/LMM pipeline

Methods & Results

15-Term DPD Mapping

HOST · basis expansion
p = 1, 3, 5; m = 0…4 → 15 terms
$x[n-m]\,|x[n-m]|^{p-1}$

LMM · basis staging
1024 samples = 512 packed pairs
2 adjacent complex samples / pair

LMM · coefficients
15 terms · 120 B
Reused across 512 pairs

PE0 Offset update → **PE1** Seed load → **PE2–PE31** 15 terms × 2 PEs Ordered accumulation → **PE32** Drain and store

One complex-MAC term · two PE stages

Real part FML / FMS → **Imaginary part** FML / FMA

Term 0 seeds the accumulator; terms 1–14 accumulate.

Accumulator stays in the pipeline until the final store.

15 terms × 2 PEs + 3 = 33 stages

Correctness & Synthetic PA Check

Check	Result	Boundary
Max. abs. error	10^-6 on IMAX	10^-7 on GPU/Jetson
NMSE change	26.1 dB	-18.75 to -44.87 dB
ACLR change	26.0 dB	-31.26 to -57.24 dBc

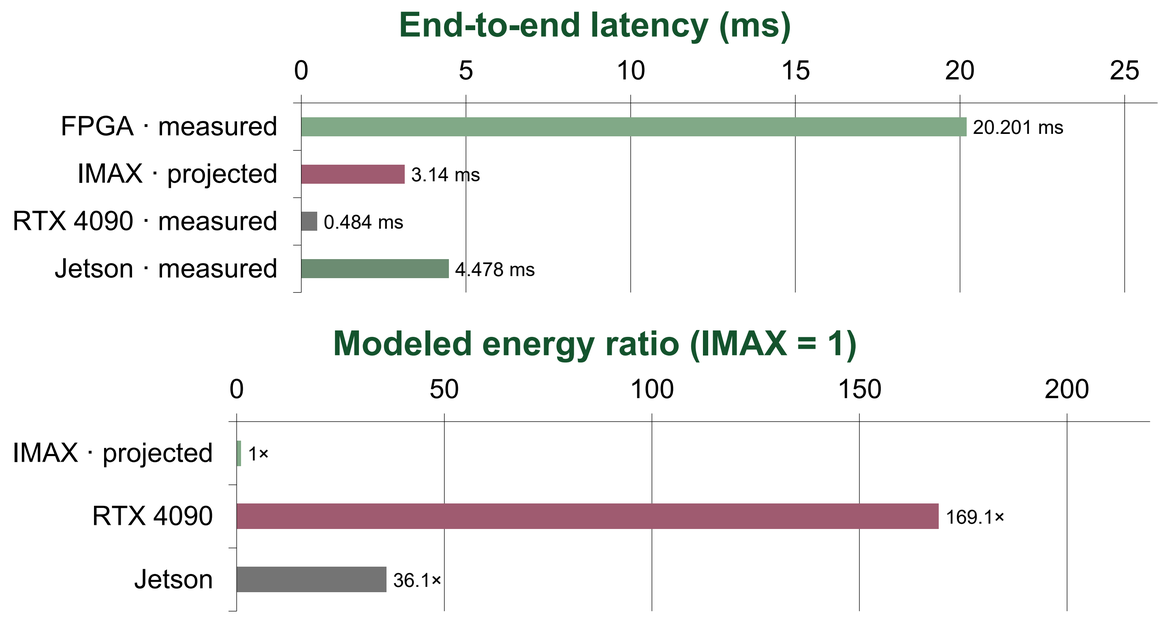
- Synthetic check validates the 15-term correction form.
- Real PA and coefficient adaptation remain outside scope.

FPGA kernel-only latency 1.948 ms
28 nm projection 0.336 ms
Host preparation remains outside the mapped reduction.

Experimental Setup

Item	Setting
Workload	32 × 2048 complex samples
Representative tile	1024 samples
FPGA	145 MHz, 64 PEs, 512 KB LMM
DPD form	(P, M) = (5, 5), 15 odd-order terms
Coefficient set	120 B

Platform Results



Top: measured/projected latency. Bottom: modeled energy under stated assumptions.

Platform	Latency type	E2E latency	Energy / batch
RTX 4090	Measured	0.484 ms	314.6 mJ*
Jetson AGX Orin	Measured	4.478 ms	67.2 mJ*
IMAX FPGA	Measured	20.201 ms	N/A
IMAX 28 nm	Projected	3.14 ms	1.86 mJ*

* GPU 650 W × 0.484 ms. Jetson 15 W × 4.478 ms.
IMAX 1.161 W × 0.336 ms + 0.525 W × 2.80 ms.
Model-based energy, not measured runtime power.

Discussion

1.86 mJ per batch | 169.1x lower modeled energy than RTX 4090*

- RTX 4090 remains the latency winner at 0.484 ms.
- Projected IMAX gives 3.14 ms under the stated model.
- FPGA E2E time is dominated by host/DMA overhead.

Conclusion

- 15-term DPD maps to a 33-stage IMAX stream.
- The result is an energy-oriented kernel mapping.
- It is not a raw-latency win over RTX 4090.

Next steps
Coefficient adaptation and continuous-streaming integration